

Summary

Dataset Overview

- **Total Records:** 7,043 customers
 - **Total Features:** 21 columns
 - **Project Objective:** Analyze customer behavior to identify the major factors contributing to customer churn and provide business recommendations to improve customer retention.
-

Customer Churn Distribution (Count Plot & Pie Chart)

Actual Results

- **Customers Retained (No Churn):** 5,174 customers (73.46%)
- **Customers Churned:** 1,869 customers (26.54%)

Interpretation

- Nearly **3 out of every 4 customers remain** with the company.
 - However, **26.54% churn** is relatively high for a telecom business and represents a significant loss of revenue.
 - The company should focus on reducing churn among high-risk customer groups.
-

Gender Distribution (Count Plot)

Actual Results

- **Male Customers:** 50.48%
- **Female Customers:** 49.52%

Interpretation

- The customer base is almost equally divided between males and females.
- Gender alone does **not significantly influence customer churn**.

- Marketing and retention strategies should target behavioral factors rather than gender.
-

Senior Citizen Analysis (Count Plot)

Actual Results

- **Non-Senior Citizens: 83.79%**
- **Senior Citizens: 16.21%**

Interpretation

- Although senior citizens represent only **16.21%** of customers, previous churn analysis indicates they leave at a higher rate than non-senior customers.
 - Senior customers may require specialized support, simplified plans, or loyalty benefits to improve retention.
-

Contract Type Analysis (Count Plot)

Actual Results

Contract Type	Percentage
Month-to-Month	55.02%
One-Year	20.91%
Two-Year	24.07%

Interpretation

- More than **half (55.02%)** of customers are on month-to-month contracts.
- Month-to-month customers are generally the most likely to churn because they can leave without long-term commitments.
- Customers with one-year or two-year contracts typically exhibit greater loyalty.

Business Recommendation

- Encourage customers to switch from month-to-month plans to annual or two-year contracts using discounts or promotional offers.

Payment Method Analysis (Count Plot)

Actual Results

Payment Method	Percentage
Electronic Check	33.58%
Mailed Check	22.89%
Bank Transfer (Automatic)	21.92%
Credit Card (Automatic)	21.61%

Interpretation

- Electronic Check is the most commonly used payment method.
 - Customers paying through Electronic Check generally have higher churn rates than those using automatic payment methods.
 - Automatic payment options help improve customer retention.
-

Internet Service Analysis (Count Plot)

Churn Rate by Internet Service

Internet Service	Retained	Churned
DSL	81.04%	18.96%
Fiber Optic	58.11%	41.89%
No Internet	92.60%	7.40%

Interpretation

- Fiber Optic customers experience the **highest churn rate (41.89%)**, more than double that of DSL users.
- Customers without internet service have the lowest churn (**7.40%**), indicating greater stability.

- Fiber Optic customers may be dissatisfied with pricing, service quality, or competition.
-

Online Security Analysis

Churn Rate

Online Security	Retained	Churned
Yes	85.39%	14.61%
No	58.23%	41.77%
No Internet Service	92.60%	7.40%

Interpretation

- Customers **without Online Security** have almost a **42% churn rate**.
- Customers using Online Security are significantly more loyal.

Business Recommendation

- Bundle Online Security with internet plans to reduce churn.
-

Tech Support Analysis

Churn Rate

Tech Support	Retained	Churned
Yes	84.83%	15.17%
No	58.36%	41.64%
No Internet Service	92.60%	7.40%

Interpretation

- Customers without Tech Support churn at **41.64%**, nearly three times the churn rate of customers with Tech Support.
- Technical assistance is one of the strongest drivers of customer retention.

Streaming TV Analysis

Churn Rate

Streaming TV	Retained	Churned
Yes	69.93%	30.07%
No	66.48%	33.52%
No Internet Service	92.60%	7.40%

Interpretation

- Streaming TV has only a modest effect on churn.
 - Customers with Streaming TV show slightly better retention than those without it.
-

Streaming Movies Analysis

Churn Rate

Streaming Movies	Retained	Churned
Yes	70.06%	29.94%
No	66.32%	33.68%
No Internet Service	92.60%	7.40%

Interpretation

- Similar to Streaming TV, Streaming Movies have only a small influence on churn.
 - Value-added entertainment services alone are insufficient to significantly reduce churn.
-

Overall Business Insights

1. **26.54%** of customers have churned, indicating a need for stronger retention strategies.

2. **55.02%** of customers are on month-to-month contracts, making this the highest-risk customer segment.
3. **Fiber Optic customers** experience the highest churn (**41.89%**), suggesting issues with pricing or service quality.
4. Customers **without Online Security** have a **41.77%** churn rate.
5. Customers **without Tech Support** have a **41.64%** churn rate.
6. Automatic payment methods are associated with better customer retention than Electronic Check.
7. Gender has little influence on churn, whereas **contract type, internet service, online security, and technical support** are the strongest predictors.

Final Conclusion

The analysis demonstrates that customer churn is primarily driven by **service-related factors rather than demographic characteristics**. The strongest predictors of churn are **month-to-month contracts, Fiber Optic internet service, lack of Online Security, and lack of Tech Support**. Focusing on long-term contracts, improving Fiber Optic customer experience, promoting security bundles, and encouraging technical support adoption can significantly improve customer retention and reduce churn.